Geographic Representation of Sequenced Hospital-Associated SARS-CoV-2 Clusters in Hong Kong: A Secondary Analysis

Shen Ruililin

Laidlaw Scholars Programme, The University of Hong Kong

Abstract

Background. Geographic patterns in genomic outbreak datasets may reflect selection into detection, sampling, sequencing, and analysis as well as underlying transmission. We assessed the geographic representation of a published sample of hospital-associated SARS-CoV-2 clusters in Hong Kong.

Methods. We reanalysed 126 genomes from 18 parent-confirmed ward clusters investigated between 28 May and 18 August 2022 [5]. The primary geographic estimand was the distribution of observed sequenced inpatients across Hospital Authority (HA) clusters relative to FY2022–23 inpatient patient-days. Beds and discharges were used as sensitivity denominators. We report both all-system representation (absent clusters contribute zero observed genomes) and represented-cluster-only concentration. Official nosocomial bulletins were used only to determine whether HA clusters absent from the genomic dataset nevertheless had reported nosocomial activity. Leave-one-ward and largest-ward exclusions assessed point influence. A multiplicative selection-bias threshold analysis translated the observed concentration into the inclusion-probability ratio that would be required under equal underlying rates. All measures describe the sequenced sample and are not estimates of nosocomial incidence.

Results. New Territories East and New Territories West accounted for 68 of 87 (78.2%) sequenced inpatients in confirmed wards. Among HA clusters represented in the genomic dataset, the New Territories-to-other-cluster sample-concentration index was approximately 4.2 using patient-days, 4.0 using beds, and 4.9 using discharges. Including absent clusters as zero sample representation raised the patient-day index to approximately 7.1. Exclusion of the largest sequenced ward cluster reduced the represented-cluster patient-day index to approximately 2.9. Kowloon Central and Hong Kong West contributed no confirmed-ward genomes despite officially reported nosocomial activity. Under a simple multiplicative selection model, an approximately 4.2-fold New Territories-to-other difference in combined inclusion probability could produce the represented-cluster pattern even if underlying rates were equal.

Conclusions. New Territories clusters constituted a larger share of the sequenced confirmed-ward inpatient sample than of aggregate inpatient throughput. Because geographic inclusion fractions were unavailable, this finding cannot be interpreted as evidence of regional differences in nosocomial incidence. Cluster-specific reported-case and genomic-inclusion denominators are required for selection-adjusted inference.

Keywords: SARS-CoV-2; genomic surveillance; geographic representation; selection bias; hospital-associated infection; Hong Kong; secondary analysis

1 Introduction

Hospital genomic outbreak datasets are selected samples. Cases must be detected, reported, sampled, successfully sequenced, and retained in analysis before they appear in a published table [1, 8, 9]. Geographic patterns in such datasets may therefore reflect the inclusion process as well as underlying transmission.

Gu et al. sequenced 162 genomes linked to 29 investigated ward units across 17 public hospitals in Hong Kong (28 May–18 August 2022), confirmed 18 nosocomial clusters (126 genomes; 51.9% of officially reported nosocomial infections in the window), and mapped cases by anonymised hospital and HA cluster [5]. The parent figure already shows geographic imbalance; Supporting Table S3 summarises official reporting [5]. What remains open is a quantitative, selection-aware description of that imbalance relative to health-system throughput, and an explicit distinction between zero sample representation and unknown disease burden.

Directional patient–staff transmission and multi-route hospital outbreaks are already well documented with richer designs than date order alone [2, 3, 7]. This note therefore does not claim to discover mixed transmission pathways. Its objective is narrower: to audit the geographic representation of the published sequenced sample.

2 Methods

2.1 Source data and cohort

We used the anonymised plot table from Gu et al. [5]. Primary analyses used the 18 parent-confirmed wards (n = 126 genomes). Confirmed clusters require at least one likely nosocomial inpatient; staff-only chains without inpatient involvement were among the excluded units in the parent study [5]. Case roles in the plot table combine staff and outpatients; we therefore label that group **non-inpatient** and do not interpret it as a single exposure category. Dates in the parent materials are sample collection dates [5]; any chronology is therefore a **first-sampling** order, not a detection or infection order.

2.2 Target estimands

We distinguish:

1. **Observed genomic representation:** sequenced confirmed-ward inpatient counts by HA cluster (zeros are true zeros for this dataset).

2. **Underlying nosocomial burden:** unknown without official hospital- or cluster-specific case totals and inclusion fractions.

3. **Sample-concentration indices (SCI):** sequenced inpatients divided by a throughput denominator (patient-days, beds, or discharges). These are not incidence rates.

2.3 Throughput standardisation

Primary denominator: FY2022–23 inpatient patient-days from HA open data [6]. Sensitivity: HA Statistical Report bed totals (31 March 2022) and FY2022–23 discharges [6]. FY2021–22 patient-days were used only as a secondary check. We compare New Territories clusters (NTEC, NTWC) with **non-New Territories** HA clusters. HA catchments are administrative service structures with cross-cluster flows and designated services [4]; throughput scaling is crude standardisation, not a closed-cohort incidence model.

We report two scopes:

- **All-system:** all seven HA clusters; absent clusters contribute 0 observed genomes.
- **Represented-cluster-only:** clusters with at least one confirmed-ward genome.

2.4 Official-presence audit

Public HA nosocomial press bulletins in the study window were coded for binary presence of each HA cluster (yes/no). Mention counts are not used as a burden proxy. Supporting Table S3 of Gu et al. remains the preferred official-total source [5]; it was not retrieved here (publisher supplement inaccessible in this environment; not embedded in the PMC HTML). Appropriate retrieval routes are journal supporting information, corresponding authors, or publisher support.

2.5 Selection-bias threshold analysis

Let Y_r be observed sequenced inpatients, D_r patient-days, I_r underlying hospital-associated inpatient infections, and s_r the combined probability of detection, reporting, sampling, successful sequencing, and inclusion in region r. Then $Y_r = s_r I_r$ and

$$R_{\mathrm{obs}} = \frac{Y_{\mathrm{NT}}/D_{\mathrm{NT}}}{Y_{\mathrm{other}}/D_{\mathrm{other}}} = R_{\mathrm{true}} \frac{s_{\mathrm{NT}}}{s_{\mathrm{other}}}. \tag{1}$$

If $R_{\mathrm{true}} = 1$, the inclusion-probability ratio required to explain R_{obs} equals R_{obs}. This does not estimate bias; it makes the identification problem quantitative.

2.6 Ward and hospital influence

We report ward and hospital counts by region, median and range of sequenced inpatients per ward, the largest ward's share, and the top-three wards' share. Leave-one-ward-out and exclusion of the largest sequenced ward recalculate the represented-cluster SCI ratio.

2.7 Secondary first-sampling analysis

As a supplement only, we classify wards by earliest non-inpatient versus inpatient sample date. This is descriptive chronology, not transmission direction [3, 7].

Table 1: Sequenced confirmed-ward persons and FY2022–23 throughput by HA cluster.

Cluster	All	IP	Non-IP	Wards	Patient-days	SCI[a]
NTWC	64	44	20	7	1,315,676	3.34
NTEC	39	24	15	5	1,349,892	1.78
KWC	12	8	4	2	1,471,927	0.54
HKEC	6	6	0	2	831,929	0.72
KEC	5	5	0	2	819,794	0.61
KCC	0	0	0	0	1,577,706	0
HKWC	0	0	0	0	608,717	0

IP: inpatient; Non-IP: staff or outpatient combined in source table. KCC/HKWC show 0 observed genomes (sample representation), not estimated zero burden.

[a] Sample-concentration index: sequenced inpatients per 100,000 FY2022–23 patient-days.

Table 2: Pooled New Territories / non-New Territories sample-concentration indices.

Denominator	Represented clusters	All-system (zeros included)
Patient-days (FY2022–23)	4.19	7.13
Beds (31 Mar 2022)	4.04	7.32
Discharges (FY2022–23)	4.87	8.42

Indices use sequenced inpatient numerators only. Not incidence rates.

3 Results

3.1 Cohort

The confirmed cohort comprised 126 sequenced persons: 87 inpatients and 39 non-inpatients across 18 wards in 13 hospitals. New Territories clusters contributed 68/87 (78.2%) sequenced inpatients and 12/18 confirmed wards (Table 1).

3.2 Geographic representation versus throughput

New Territories clusters held 78.2% of sequenced confirmed-ward inpatients but a much smaller share of system patient-days (Figure 1). Among represented clusters, the NT/other SCI ratio was approximately 4.2 (patient-days), 4.0 (beds), and 4.9 (discharges) (Figure 2; Table 2). Treating absent clusters as zero sample representation raised the patient-day ratio to approximately 7.1 and the bed ratio to approximately 7.3. These two scopes answer different questions: represented-cluster concentration versus all-system geographic representation.

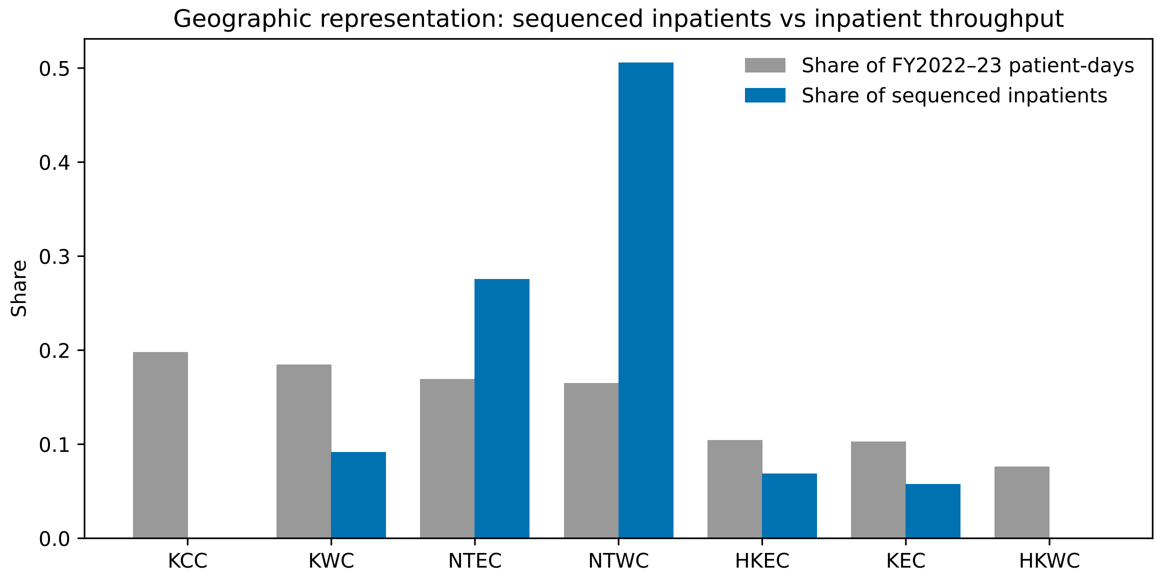

Figure 1: Share of sequenced confirmed-ward inpatients versus share of FY2022–23 inpatient patient-days by HA cluster.

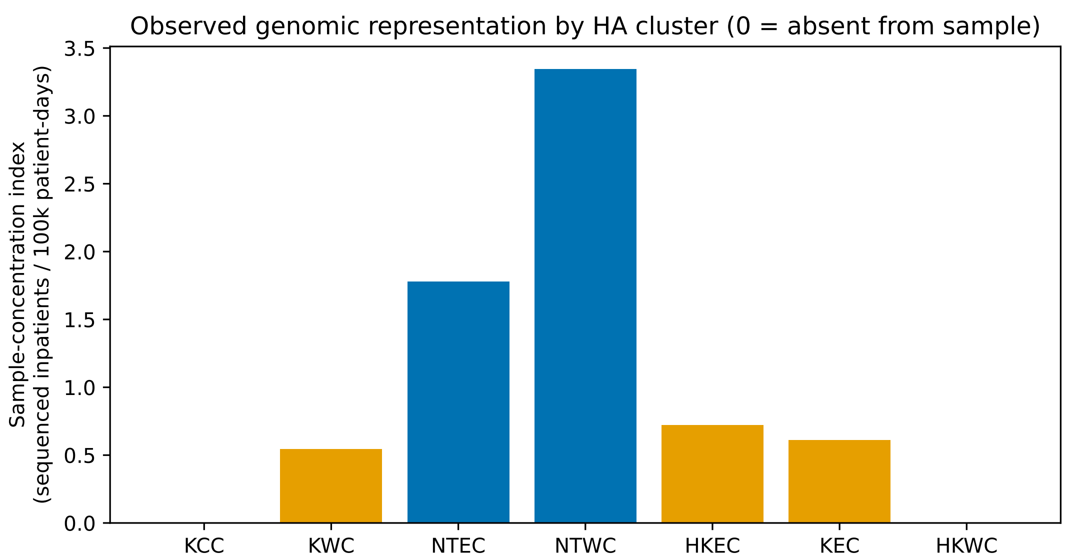

Figure 2: Sample-concentration index by HA cluster (sequenced inpatients per 100,000 patient-days). Zeros denote absence from the genomic sample.

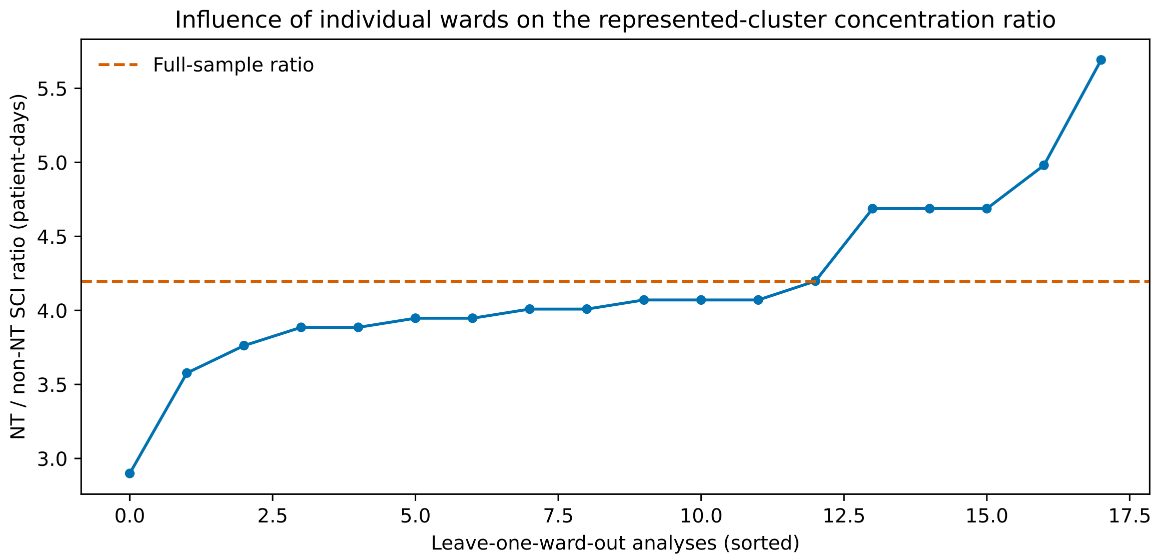

Figure 3: Leave-one-ward-out New Territories / non-New Territories sample-concentration ratios (patient-days, represented clusters).

Table 3: Binary audit of official bulletin presence versus genomic-sample presence.

HA cluster	Official nosocomial bulletin in window	Confirmed-ward genome in dataset
HKEC	Yes	Yes
HKWC	Yes	No
KCC	Yes	No
KEC	Yes	Yes
KWC	Yes	Yes
NTEC	Yes	Yes
NTWC	Yes	Yes

3.3 Ward influence

Sequenced inpatients per ward ranged from 2 to 21 (median 3.5). The largest ward contributed 24.1% of sequenced inpatients; the top three wards contributed 43.7%. Leave-one-ward-out represented-cluster patient-day ratios ranged from approximately 2.9 to 5.7 (median 4.0; Figure 3). Excluding the largest sequenced ward reduced the represented-cluster patient-day ratio to approximately 2.9 and the bed ratio to approximately 2.8. Concentration is therefore point-influenced but not created solely by one ward.

3.4 Official presence versus genomic presence

All seven HA clusters appeared in official nosocomial bulletins during the window, including Kowloon Central and Hong Kong West, which contribute zero confirmed-ward genomes to the analytic dataset (Figure 4; Table 3). Absence from the genomic table is therefore not absence from official reporting.

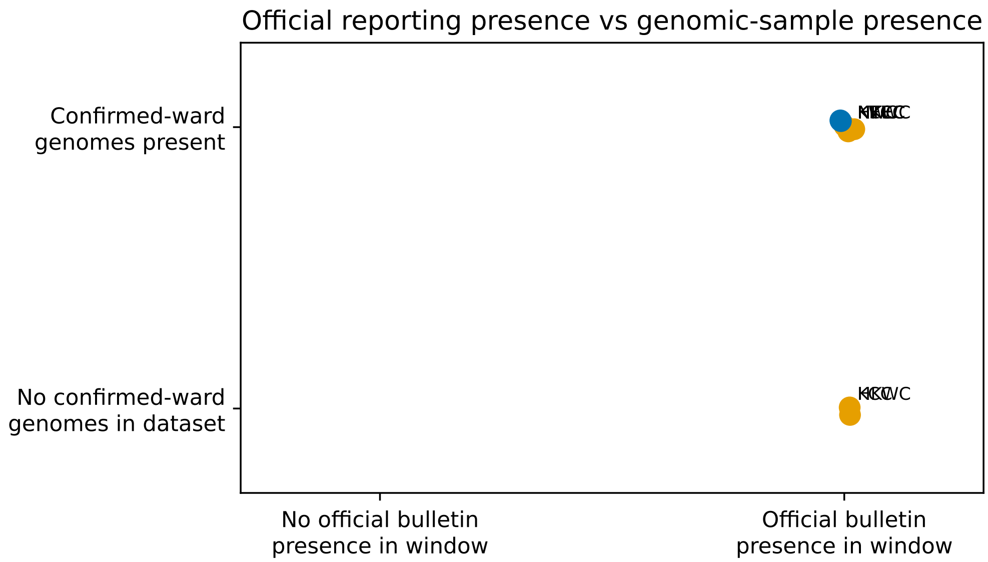

Figure 4: Official reporting presence versus presence of confirmed-ward genomes in the analytic dataset.

3.5 Selection-bias threshold

For the represented-cluster patient-day SCI ratio $R_{\text{obs}} \approx 4.2$, equation (1) implies that equal underlying rates ($R_{\text{true}} = 1$) would require $s_{\text{NT}}/s_{\text{other}} \approx 4.2$ (Figure 5). Larger inclusion ratios would imply $R_{\text{true}} < 1$; smaller ratios would leave residual true-rate differences. Without cluster-specific inclusion fractions, R_{true} is not identified.

3.6 Secondary first-sampling chronology

Among confirmed wards, both non-inpatient-before-inpatient and inpatient-before-non-inpatient first-sample orders occurred (supplement). Because confirmation requires an inpatient, non-inpatient-only pathways are confined to non-confirmed units by definition and are not interpreted as an independent finding [5]. First-sampling order is not infection order, index status, or transmission direction [3, 7].

4 Discussion

The sequenced confirmed-ward inpatient sample is geographically concentrated in New Territories clusters relative to crude inpatient throughput. That concentration survives alternative denominators and leave-one-ward checks, shrinks when the largest ward is excluded, and coexists with official nosocomial reporting from clusters absent from the genomic table. The appropriate interpretation is selection-aware sample geography, not regional incidence [8].

Hospital genomic epidemiology already shows multi-route transmission with designs far stronger than sampling-date order [2, 3, 7]. Prospective sequencing programmes also show incomplete return of sequence

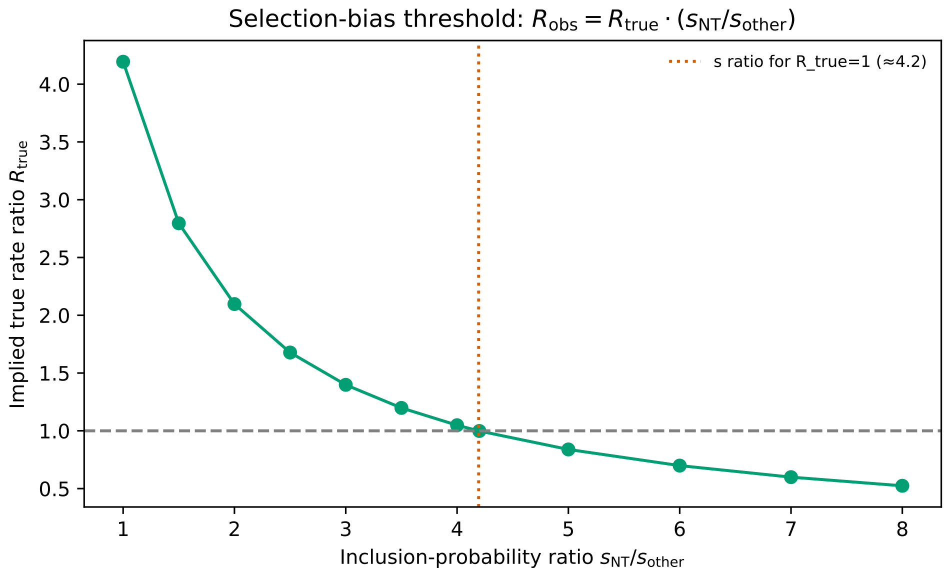

Figure 5: Selection-bias threshold analysis under $R_{obs} = R_{true} \cdot (s_{NT}/s_{other})$.

reports and limited evidence that sequencing alone reduces hospital incidence [9]. Against that literature, the contribution of this note is corrective: throughput-standardised quantification of imbalance in one published Hong Kong sample, plus an explicit selection threshold.

Ward function, length of stay, testing intensity, and service configuration may affect both outbreak size and inclusion in genomic investigation; these factors were not measured here and are not attributed to any specialty group.

4.1 Limitations

1. Sequenced confirmed-ward cases are not complete nosocomial incidence.
2. Annual patient-days imperfectly proxy an 83-day window.
3. Non-inpatient combines staff and outpatients.
4. Official bulletin presence is not a quantitative burden measure; Supporting Table S3 was not retrieved.
5. Hospital-letter denominator linkage is omitted from the main text to reduce re-identification risk.
6. HA catchments are not closed populations [4].

5 Conclusion

New Territories clusters constituted a larger share of the sequenced confirmed-ward inpatient sample than of aggregate inpatient throughput. Because geographic inclusion fractions were unavailable, this pattern cannot be read as regional differences in nosocomial incidence. Cluster-specific reported-case and genomic-inclusion denominators are required for selection-adjusted inference.

Data and code

Primary script: `scripts/8.geographic_representation.py`. Outputs: `analysis/outputs/geo_*`; first-sampling supplement: `manuscript/supplement/`. Parent study: Gu et al. [5].

Competing interests

None declared.

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